Airline Operations Report

The report summarizes flight operations, including key performance metrics such as on-time performance, cancellations, delays, flight volumes, and passenger load across the airlines and destinations served. It highlights the new Airline App that provides detailed analytics, real-time insights, and a user-friendly interface for tracking all flight operations.

Key Performance Metrics

- Total Tickets Booked- 50
- Total Airlines- 4
- Total Operating Flights-78
- Passenger Load-100
- On-time Flight Rate- 53.85%
- Cancellations Rate- 29.49%
- Delayed Flight Rate- 16.66%
- Total Destinations Covered- 5

Destinations

- Phoenix
- Houston
- Chicago
- New York
- Los Angeles

Airline App Features

It has been developed to provide detailed and real-time insights into airline operations and flight performance. It includes the following key features:

- On-time Flight Monitoring
- Airline Statistics
- Top Influencers
- Q&A Preview
- Destination-wise Flight Classification
- Key Insights
- Airline Operations Monitoring
- Interactive Dashboards

Airline app-
https://app.powerbi.com/Redirect?action=OpenApp&appId=b9d7508e-51f6-4288-b58f-ba5331720071&ctid=276e0cb4-cbcf-4a5a-8266-cd933c139077

Embed link for SharePoint-
https://app.powerbi.com/reportEmbed?reportId=d757724b-ca51-4375-b01e-3c959486c439&appId=b9d7508e-51f6-4288-b58f-ba5331720071

Airline and Flight Performance

- *Airline A* is the top-performing airline regarding the number of flights operated, with 22 flights.
- *Phoenix* is the busiest destination.
- *Phoenix* and *Houston* were the most serviced destinations, with 20 flights each.
- *Chicago* had the highest number of cancellations with 7 cancelled flights, indicating potential operational challenges at this destination.

Flight Performance by Airline

S. No	Airline	Flights	Passengers	On-Time Performance (%)	Cancellation Rate (%)	Delayed Flight Rate (%)
1	Airline A	22	30	45.45	36.36	18.14
2	Airline B	15	20	49.67	20	33.33
3	Airline C	18	22	50	38.89	11.11
4	Airline D	26	28	42.31	19.23	38.46

Airline-Specific Observations and Suggestions

- **Airline A**: Despite operating the highest number of flights, Airline A's operational performance is hindered by its high cancellation rate and low on-time performance. Management must track flight disruptions and their patterns, addressing the issues associated with cancellations and delays with the help of the airline app.

- **Airline B**: Its solid on-time performance is a strong point. However, its delayed flight rate suggests there is room for improvement in minimizing delays. The app's real-time tracking and key insights section can help identify areas for improving efficiency.

- **Airline C**: It experiences an average on-time performance of 50%, but it struggles with a high cancellation rate. Focus on reducing the high cancellation rate with the help of Destination-wise Flight Classification. The main area of concern is cancellation which has to be addressed timely to improve the efficiency of the airline.

- **Airline D**: It has the highest delayed flight rate at 38.46%. It needs immediate attention regarding its delays, which are significantly higher than the other airlines. Investigate and address the high delayed flight rate with data insights from the app, focusing on improving operations across all flights.

NOTE:

1. Calculations were done for analysis using DAX Expressions.

a) Total Flights = DISTINCTCOUNT(Flight_Passenger[FlightNumber])

*b) Cancelled rate = DIVIDE(Flight_Passenger[Total Cancelled flights],'Fight_Information - flight_information'[Total Flights])*100*

*c) On-time rate = DIVIDE(Flight_Passenger[Total Ontime flights],'Fight_Information - flight_information'[Total Flights])*100*

d) Number of Flights per Airline = CALCULATE(DISTINCTCOUNT('Flight_Passenger'[FlightID]),ALLEXCEPT('Flig h t_Passenger', 'Flight_Passenger'[Airline]))

e) Total Cancelled flights = CALCULATE(COUNTROWS('Flight_Passenger'),Flight_Passenger[Status] = "Cancelled")

f) Total Ontime flights = CALCULATE(COUNTROWS('Flight_Passenger'),Flight_Passenger[Status] = "On-Time")

g) Total Passengers per Airline = CALCULATE(DISTINCTCOUNT(Flight_Passenger[PassengerID]),ALLEXCEPT(Fl i ght_Passenger, Flight_Passenger[Airline]))

h) Total Passengers for FL1955 = CALCULATE(COUNTROWS('Fight_Information - flight_information'),'Fight_Information - flight_information'[FlightNumber] = "FL1955")

i) Total Tickets Booked = COUNTROWS('Ticket_Information - ticket_information')

j) Best Flights = CALCULATETABLE('Flight_Passenger', Flight_Passenger[Performance] = "Best")

2. A New Table was created Flight_Passenger for better analysis.